



Practice Programs

- By Aditya Varma

Index

	Program Question
Program 1:	Printing a Pun
Program 2:	Simple Arithmetic
Program 3:	Simple Interest Calculator
Program 4:	Simple EMI Calculator
Program 5:	Student Percentage Calculator.
Program 6:	Gross Salary Calculator
Program 7:	Calculate Area of Shapes
Program 8:	Calculate Area & Circumference of Circle
Program 9:	Calculate Dimensional Weight of a Box
Program 10:	Convert Celsius into Fahrenheit
Program 11:	Convert Fahrenheit into Celsius
Program 12:	Convert Celsius into Kelvin
Program 13:	Convert Kelvin into Celsius
Program 14:	Convert Distance into other Measures
Program 15:	Bitwise Operator
Program 16:	Increment Operator
Program 17:	Decrement Operator
Program 18:	sizeof Operator
Program 19:	Find ASCII Value
Program 20:	Print Number in Format
Program 21:	Add Fractions
Program 22:	UPC Code Checker
Program 23:	Student Marksheet
Program 24:	Trigonometry
Program 25:	Logarithm
Program 26:	Type of Character Input
Program 27:	Leap Year or Not
Program 28:	Leap Year or Not (Ternary Operator)
Program 29:	Greatest Number
Program 30:	Broker's Commission
Program 31:	Digit to words
Program 32:	Simple Calculator
Program 33:	Date Output
Program 34:	Automorphic
Program 35:	Linear Search

Program Question	
Program 36:	Perfect Number
Program 37:	Prime Number
Program 38:	Multiplication Table
Program 39:	Reverse a Number
Program 40:	Reverse a String
Program 41:	Star Pattern
Program 42:	Star Pattern 2
Program 43:	Number Pattern
Program 44:	Number Pattern 2
Program 45:	Alphabet Pattern
Program 46:	Alphabet Pattern 2
Program 47:	Alphabet-Number Pattern
Program 48:	Sum of Integers
Program 49:	Message Length
Program 50:	Check book
Program 51:	Reverse Series of Number
Program 52:	Repeated Digit
Program 53:	Random Cards
Program 54:	Interest Table
Program 55:	Pun Print Function
Program 56:	Average of Number
Program 57:	Countdown
Program 58:	Prime Number Function
Program 59:	Factorial
Program 60:	Armstrong
Program 61:	Fibonacci
Program 62:	Quick Sort
Program 63:	Banking Application
Program 64:	Tic Tac Toe Game
Program 65:	Library Application

Program 1: Write a C program to print the following text message each time it is run.
'To C, or not to C: that is the question.'

pun.c:

```
1  #include <stdio.h>
2
3  int main() {
4      printf("To C, or not to C: that is the question.\n");
5      return 0;
6  }
```

Program 2: Write a C program to demonstrate simple arithmetic.

simpleArithmetic.c

```
1  #include<stdio.h>
2
3  int main(){
4      int a, b, c;
5
6      printf("Enter two numbers (eg.:5 8): ");
7      scanf("%d %d", &a, &b);
8
9      c = a + b; // addition
10     printf("The Sum is %d\n", c);
11
12     c = a - b; // subtraction
13     printf("The Difference is %d\n", c);
14
15     c = a * b; // multiplication
16     printf("The Product is %d\n", c);
17
18     c = a / b; // division
19     printf("The Division is %d\n", c);
20
21     return 0;
22 }
```


Program 3: Write a C program to calculate Simple Interest.**simpleInterestCalculator.c**

```

1  #include<stdio.h>
2
3  int main(){
4      int numberOfYear;
5      float principleAmount, rateOfInterest, simpleInterest;
6
7      printf("Enter Principle Amount: ");
8      scanf("%f", &principleAmount);
9      printf("Enter number of year: ");
10     scanf("%d", &numberOfYear);
11     printf("Enter Rate of Interest: ");
12     scanf("%f", &rateOfInterest);
13
14     simpleInterest = (principleAmount * numberOfYear * rateOfInterest) / 100;
15     printf("Simple interest is %f\n", simpleInterest);
16
17     return 0;
18 }

```

Program 4: Write a C program to calculate Simple EMI.**simpleEMICalculator.c**

```

1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      float principle, rate, time, emi;
6
7      printf("Enter principle amount: ");
8      scanf("%f", &principle);
9
10     printf("Enter rate of interest: ");
11     scanf("%f", &rate);
12
13     printf("Enter time in years: ");
14     scanf("%f", &time);
15
16     rate = rate / (12 * 100); //one month interest
17     time = time * 12; //one month period
18
19     emi = (principle * rate * pow(1+rate, time)) / (pow(1+rate,time) - 1);
20
21     printf("\nMonthly EMI is= %f\n", emi);
22
23     return 0;
24 }

```

Program 5: Write a C program to calculate Student Percentage of 5 subjects

studentPercentageCalculator.c

```

1  #include <stdio.h>
2
3  int main() {
4      int a, b, c, d, e;
5      float total, average, percentage;
6
7      printf("Enter marks of 5 subjects: ");
8      scanf("%d %d %d %d %d", &a, &b, &c, &d, &e);
9
10     total = a + b + c + d + e;
11     average = total / 5;
12     percentage = (total / 500) * 100;
13
14     printf("Total marks = %f\n", total);
15     printf("Average marks = %f\n", average);
16     printf("Net percentage = %f\n", percentage);
17
18     return 0;
19 }
```

Program 6: Write a program taking user input of basic salary and calculating gross salary that includes basic salary, 50% DA and 40% HRA.

grossSalaryCalculator.c

```

1  #include <stdio.h>
2
3  int main() {
4      float basic, hra, da, gross;
5
6      printf("Enter the Basic Salary: $");
7      scanf("%f", &basic);
8
9      hra = 40 * basic / 100;
10     da = 50 * basic / 100;
11     gross = basic + hra + da;
12
13     printf("Gross Salary is $%f", gross);
14
15     return 0;
16 }
```

Program 7: Write a program to calculate area of Square, Triangle & Rectangle using user inputs of shape data.

areaOfShapes.c

```

1  #include <stdio.h>
2
3  int main() {
4      float area;
5      float side; // For square
6      float base, height; // For triangle
7      float length, breadth; // For rectangle
8
9      // Area of square
10     printf("Enter the side of square in cms: ");
11     scanf("%f", &side);
12
13     area = side * side;
14     printf("Area of square with sides %.2f cms is %f\n", side, area);
15
16     // Area of Triangle
17     printf("Enter the base of triangle in cms: ");
18     scanf("%f", &base);
19     printf("Enter the height of triangle in cms: ");
20     scanf("%f", &height);
21
22     area = 0.5 * base * height;
23     printf("Area is of triangle with base %.2fcms and height %.2fcms is %fcms\n", base, height, area);
24
25     // Area of Rectangle
26     printf("Enter the length of rectangle in cms: ");
27     scanf("%f", &length);
28     printf("Enter the breadth of rectangle in cms: ");
29     scanf("%f", &breadth);
30
31     area = length * breadth;
32     printf("The area of rectangle with length %.2fcms and breadth %.2fcms is %fcms\n", length, breadth, area);
33
34     return 0;
35 }

```

Program 8: Write a program to calculate Area & Circumference of a Circle, take user inputs.

areaAndCircumference.c

```

1  #include <stdio.h>
2
3  int main() {
4      float radius, area, circumference;
5
6      printf("Enter the radius of Circle in cms: ");
7      scanf("%f", &radius);
8
9      area = 3.14 * radius * radius;    // Area of circle
10     circumference = 2 * 3.14 * radius; // Circumference of circle
11
12     printf("Area of circle is %f\n", area);
13     printf("Circumference of circle is %f\n", circumference);
14
15     return 0;
16 }

```

Program 9: Write a program to calculate Dimensional weight of a box to help shipping companies charge accordingly.

dimensionalWeight.c

```

1  /*The problem is about calculating the dimensional weight of a box,
2   which shipping companies use to charge based on space taken rather
3   than actual weight. The formula divides the box's volume by 166,
4   but since integer division in C truncates decimals (rounds down),
5   we adjust by adding 165 before dividing to properly round up.*/
6
7  #include <stdio.h>
8
9  int main() {
10     int height, length, width, volume, weight;
11
12     printf("Enter height of the box: ");
13     scanf("%d", &height);
14     printf("Enter length of the box: ");
15     scanf("%d", &length);
16     printf("Enter width of the box: ");
17     scanf("%d", &width);
18
19     volume = height * length * width;
20     weight = (volume + 165) / 166;
21
22     printf("Volume of box is %d (cubic meters)\n", volume);
23     printf("Dimensional weight of the box is %d kgs\n", weight);
24
25     return 0;
26 }

```

Program 10: Write a program to convert the user given temperature in Celsius(*C) into Fahrenheit(*F).

celsiusToFahrenheit.c

```
1  #include <stdio.h>
2
3  #define FREEZING_PT 32.0f
4  #define SCALE_FACTOR (9.0f / 5.0f)
5
6  int main() {
7      float celsius, fahrenheit;
8
9      printf("Enter the Temperature in Celcius: ");
10     scanf("%f", &celsius);
11
12     fahrenheit = SCALE_FACTOR * celsius + FREEZING_PT;
13
14     printf("Temperature in Fahernheit is %.1f F", fahrenheit);
15
16     return 0;
17 }
```

Program 11: Write a program to convert the user given temperature in Fahrenheit(*F) into Celsius(*C).

fahrenheitToCelsius.c

```
1  #include <stdio.h>
2
3  #define FREEZING_PT 32.0f
4  #define SCALE_FACTOR (5.0f / 9.0f)
5
6  int main() {
7      float fahrenheit, celsius;
8
9      printf("Enter the Temperatue in Fahrenheit: ");
10     scanf("%f", &fahrenheit);
11
12     celsius = (fahrenheit - FREEZING_PT) * SCALE_FACTOR;
13
14     printf("Temperature in Celsius is %.1f\n C", celsius);
15
16     return 0;
17 }
```


Program 12: Write a program to convert the user given temperature in Celsius(*C) to Kelvin(*K).

celsiusToKelvin.c

```

1  #include <stdio.h>
2
3  #define SCALE_FACTOR 273.15
4
5  int main() {
6      float celsius, kelvin;
7
8      printf("Enter the Temperature in Celcius: ");
9      scanf("%f", &celsius);
10
11     kelvin = celsius + SCALE_FACTOR;
12
13     printf("Temperature in Kelvin is %.2f k\n", kelvin);
14
15     return 0;
16 }
```

Program 13: Write a program to convert the user given temperature in Kelvin(*K) into Celsius(*C).

kelvinToCelsius.c

```

1  #include <stdio.h>
2
3  #define SCALE_FACTOR 273.15
4
5  int main() {
6      float celsius, kelvin;
7
8      printf("Enter the Temperature in Kelvin: ");
9      scanf("%f", &kelvin);
10
11     celsius = kelvin - SCALE_FACTOR;
12
13     printf("Temperature in Celsius is %.2f C\n", celsius);
14
15     return 0;
16 }
```

Program 14: Write a program to accept the distance between two cities in kilometres from the user. Calculate and display this distance in meters, feet, centimetres and inches

distanceConverter.c

```

1  #include <stdio.h>
2  #include <conio.h>
3
4  int main() {
5      float km, mt, inch, ft, cm;
6
7      printf("Enter the distance between two cities in kilometers: ");
8      scanf("%f", &km);
9
10     mt = km * 1000;
11     ft = mt * 3.33;
12     cm = mt * 100;
13     inch = ft * 12;
14
15     printf("The distance in meters is = %.2f mts.\n", mt);
16     printf("The distance in feets is = %.2f ft.\n", ft);
17     printf("The distance in centimeters is = %.2f cms.\n", cm);
18     printf("The distance in inchs is = %.2f inches.\n", inch);
19
20     return 0;
21 }
```

Program 15: Write a program to demonstrate bitwise operator on two integers.

bitwiseCalculation.c

```

1  #include <stdio.h>
2
3  int main() {
4      int a = 14, b = 7;
5
6      printf("Binary of %d is 00001100\n", a);
7      printf("Binary of %d is 00000111\n", b);
8
9      printf("Bitwise NOT of a (~a): %d\n", ~a);
10     printf("Bitwise NOT of b (~b): %d\n\n", ~b);
11
12     printf("a & b = %d\n", a & b);
13     printf("a | b = %d\n", a | b);
14     printf("a ^ b = %d\n\n", a ^ b);
15
16     printf("a << 1 = %d\n", a << 1);
17     printf("b << 1 = %d\n\n", b << 1);
18
19     printf("a >> 1 = %d\n", a >> 1);
20     printf("b >> 1 = %d\n", b >> 1);
21
22     return 0;
23 }
```

Program 16: Write a program to demonstrate increment operator on an integer.

incrementOperator.c

```

1 // Increment operator
2
3 #include<stdio.h>
4 int main()
5 {
6     int a= 5;
7     int b, c, d;
8
9     printf("Value of a = %d\n", a);
10
11     b = ++a;
12     c = a++;
13     d = ++a;
14
15     printf("Value of b using ++a = %d\n", b);
16     printf("Value of c using a++ = %d\n", c);
17     printf("Value of d using ++a = %d\n", d);
18
19     return 0;
20 }
```

Program 17: Write a program to demonstrate decrement operator on an integer.

decrementOperator.c

```

1 #include <stdio.h>
2
3 int main() {
4     int a = 8;
5     int b, c, d;
6
7     printf("Value of a = %d\n", a);
8
9     b = --a;
10    c = a--;
11    d = --a;
12
13    printf("Value of b using --a = %d\n", b);
14    printf("Value of c using a-- = %d\n", c);
15    printf("Value of d using --a = %d\n", d);
16
17    return 0;
18 }
```


Program 18: Write a program to calculate size of variables.

calculateSizeOfVariables.c

```

1  /* sizeof() operator in c is used to calculate the
2     size of a variable used inside the program and
3     return the size in integer in the form of
4     memory bytes*/
5
6  #include <stdio.h>
7
8  int main() {
9      int a = 6;
10     float b = 8.765f;
11     long long int c = 15050603LL;
12     double d = 78945.321654;
13     char ch = 'X';    // Capital X
14
15     printf("Size of a : %d Bytes\n", sizeof(a));
16     printf("Size of b : %d Bytes\n", sizeof(b));
17     printf("Size of c : %d Bytes\n", sizeof(c));
18     printf("Size of d : %d Bytes\n", sizeof(d));
19     printf("Size of X : %d Bytes\n", sizeof(ch));
20
21     return 0;
22 }
```

Program 19: Write a program to find ASCII value of the character user inputs.

findASCIIValue.c

```

1  #include <stdio.h>
2
3  int main() {
4      char c;
5
6      printf("Enter a character: ");
7
8      // Reads character input from the user
9      scanf("%c", &c);
10
11     // %d displays the integer value of a character
12     // %c displays the actual character
13     printf("ASCII value of %c = %d", c, c);
14
15     return 0;
16 }
```

Program 20: Write a program to output numbers in certain formats using printf() function.

formattedOutput.c

```

1  #include <stdio.h>
2
3  int main(void) {
4      int i;
5      float x;
6
7      i = 40;
8      x = 839.21f;
9
10     printf("|%d|%5d|%-5d|%5.3d|\n", i, i, i, i);
11     printf("|%10.3f|%10.3e|%-10g|\n", x, x, x);
12
13     return 0;
14 }
```

Program 21: Write a program to add two fractions taken from the user and show the result.

addTwoFraction.c

```

1  #include <stdio.h>
2
3  int main() {
4      int num1, denom1, num2, denom2, result_num, result_denom;
5
6      printf("Enter first fraction(a/b): ");
7      scanf("%d/%d", &num1, &denom1);
8
9      printf("Enter second fraction(c/d): ");
10     scanf("%d/%d", &num2, &denom2);
11
12     result_num = num1 * denom2 + num2 * denom1;
13     result_denom = denom1 * denom2;
14
15     printf("The sum is %d/%d\n", result_num, result_denom);
16
17     return 0;
18 }
```

Program 22: Write a program to add two fractions taken from the user and show the result.

upcCodeCheck.c

```

1  #include <stdio.h>
2
3  int main() {
4      int d, i1, i2, i3, i4, i5, j1, j2, j3, j4, j5;
5      int first_sum, second_sum, total;
6
7      printf("Enter the first (single) digit: ");
8      scanf("%d", &d);
9      printf("Enter first group of five digits: ");
10     scanf("%d%d%d%d%d", &i1, &i2, &i3, &i4, &i5);
11     printf("Enter second group of five digits: ");
12     scanf("%d%d%d%d%d", &j1, &j2, &j3, &j4, &j5);
13
14     first_sum = d + i2 + i4 + j1 + j3 + j5;
15     second_sum = i1 + i3 + i5 + j2 + j4;
16     total = 3 * first_sum + second_sum;
17
18     printf("Check digit: %d\n", 9 - ((total - 1) % 10));
19
20     return 0;
21 }
```

Program 23: Write a program to display the marksheet of the students.

studentMarksheet.c

```

1  /*Evaluate total, average and percentage of a student*/
2
3  #include<stdio.h>
4
5  int main() {
6      int a,b,c,d,e;
7      float total,average,percentage;
8
9      printf("Enter marks of 5 subjects : \n");
10     scanf("%d %d %d %d %d",&a,&b,&c,&d,&e);
11
12     total = a+b+c+d+e;
13     average = total/5;
14     percentage = (total/500)*100;
15
16     printf("Total marks = %f\n", total);
17     printf("Average marks = %f\n",average );
18     printf("Net percentage = %f\n",percentage);
19     return 0;
20 }
```

Program 24: Write a program to calculate tan angle of a given angle.

tanAngle.c

```

1  #include <stdio.h>
2  #include <math.h>
3
4  void main() {
5      double a, b;
6
7      printf("Enter the degree: ");
8
9      scanf("%lf", &a);
10
11     a = (a * 3.14) / 180;
12     b = tan(a);
13
14     printf("tan is %lf", b);
15 }
```

Program 25: Write a program to calculate log value of a given number.

logarithm.c

```

1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      double value, result;
6
7      printf("Enter the value of Log to calculate: ");
8      scanf("%lf", &value);
9
10     // Calculate the log of the value
11     result = log(value);
12
13     // Display the result of the calculation
14     printf("The Natural Logarithm of %lf is %lf\n", value, result);
15
16     return 0;
17 }
```

Program 26: Write a program to check the type of input character given by user.

checkCharacterInput.c

```

1  #include <stdio.h>
2
3  int main() {
4      char ch;
5
6      printf("Enter a character: ");
7      scanf("%c", &ch);
8
9      if((ch >= 65 && ch <= 90) || (ch >= 97 && ch <= 122)) {
10         printf("Entered character is an alphabet");
11     } else if (ch >= 48 && ch <= 57) {
12         printf("Entered character is a digit");
13     } else {
14         printf("Entered character is a special symbol");
15     }
16
17     return 0;
18 }
```

Program 27: Write a program to check if the year given by user is a leap year or not.

isLeapYear.c

```

1  #include <stdio.h>
2
3  int main() {
4      int input, i;
5
6      printf("Please input a year and I will tell you if it's a leap year: ");
7      scanf("%d", &input);
8
9      if((input % 4 == 0 && input % 100 != 0) || input % 400 == 0) {
10         printf("Your input (%d) IS a leap year.\n\n", input);
11     } else {
12         printf("Your input (%d) is NOT a leap year.\n\n", input);
13     }
14
15     return 0;
16 }
```


Program 28: Write a program to check if the year given by user is a leap year or not using ternary operator.

isLeapYearTernary.c

```

1  #include <stdio.h>
2
3  int main() {
4      int y;
5
6      printf("Please input a year and I will tell you if it's a leap year: ");
7      scanf("%d",&y);
8
9      //Using ternary operator
10     ((y % 100 != 0) && (y % 4 == 0)) || (y % 400 == 0) ?
11         printf("Your input (%d) is a leap year.",y) :
12         printf("Your input (%d) is NOT a leap year.",y);
13
14     return 0;
15 }
```

Program 29: Write a program to check the greatest number given by user.

greatestNumber.c

```

1  #include <stdio.h>
2
3  int main() {
4      int a, b, c;
5
6      printf("Enter three integer number: ");
7      scanf("%d %d %d", &a, &b, &c);
8
9      if(a>b) {
10         if(a>c)
11             printf("%d is greatest", a);
12         else
13             printf("%d is greatest", c);
14     } else {
15         if(b>c)
16             printf("%d is greatest", b);
17         else
18             printf("%d is greatest", c);
19     }
20
21     return 0;
22 }
```


Program 30: Write a program to calculate the commission of broker on a trade value.

brokersCommission.c

```
1  #include <stdio.h>
2
3  int main() {
4      float commission, value;
5
6      printf("Enter value of trade: $");
7      scanf("%f", &value);
8
9      if (value < 2500.00){
10         commission = 30.00 + .017 * value;
11     } else if (value < 6250.00) {
12         commission = 56.00 + .0066 * value;
13     } else if (value < 20000.00) {
14         commission = 76.00 + .0034 * value;
15     } else if (value < 50000.00) {
16         commission = 100.00 + .0022 * value;
17     } else if (value < 500000.00) {
18         commission = 155.00 + .0011 * value;
19     } else {
20         commission = 255.00 + .0009 * value;
21     }
22
23     if (commission < 39.00){
24         commission = 39.00;
25     }
26
27     printf("Commission: $%.2f\n", commission);
28
29     return 0;
30 }
```

Program 31: Write a program to output a digit given by user in words.

digitToWords.c

```

1  #include <stdio.h>
2
3  int main() {
4      int digit;
5
6      printf("Enter a single digit number (0-9): ");
7      scanf("%d", &digit);
8
9      switch (digit) {
10         case 0:
11             printf("Zero\n");
12             break;
13         case 1:
14             printf("One\n");
15             break;
16         case 2:
17             printf("Two\n");
18             break;
19         case 3:
20             printf("Three\n");
21             break;
22         case 4:
23             printf("Four\n");
24             break;
25         case 5:
26             printf("Five\n");
27             break;
28         case 6:
29             printf("Six\n");
30             break;
31         case 7:
32             printf("Seven\n");
33             break;
34         case 8:
35             printf("Eight\n");
36             break;
37         case 9:
38             printf("Nine\n");
39             break;
40         default:
41             printf("Invalid input! Please enter a single digit (0-9).\n");
42     }
43
44     return 0;
45 }

```

Program 32: Write a program to give simple calculator options to user.

simpleCalculator.c

```

1  #include <stdio.h>
2
3  int main() {
4      int choice;
5      float num1, num2, result;
6
7      printf("==== Simple Calculator ==== \n");
8      printf("1. Addition \n");
9      printf("2. Subtraction \n");
10     printf("3. Multiplication \n");
11     printf("4. Division \n");
12     printf("5. Modulus \n");
13     printf("Enter your choice (1-5): ");
14     scanf("%d", &choice);
15
16     // For modulus, we'll use int inputs, so read accordingly
17     if (choice >= 1 && choice <= 5) {
18         printf("Enter two numbers: ");
19         scanf("%f %f", &num1, &num2);
20     }
21
22     switch (choice) {
23     case 1:
24         result = num1 + num2;
25         printf("Result = %.2f \n", result);
26         break;
27     case 2:
28         result = num1 - num2;
29         printf("Result = %.2f \n", result);
30         break;
31     case 3:
32         result = num1 * num2;
33         printf("Result = %.2f \n", result);
34         break;
35     case 4:
36         if (num2 != 0)
37             printf("Result = %.2f \n", num1 / num2);
38         else
39             printf("Error! Division by zero not allowed. \n");
40         break;
41     case 5:
42         // Modulus only makes sense for integers
43         printf("Result = %d \n", (int)num1 % (int)num2);
44         break;
45     default:
46         printf("Invalid choice! Please enter between 1 and 5. \n");
47     }
48
49     return 0;
50 }

```

Program 33: Write a program to output date in legal form.

date.c

```

1  #include <stdio.h>
2
3  int main() {
4      int month, day, year;
5
6      printf("Enter date (mm/dd/yy): ");
7      scanf("%d /%d /%d", &month, &day, &year);
8
9      printf("Dated this %d", day);
10
11     switch (day) {
12         case 1: case 21: case 31:
13             printf("st"); break;
14         case 2: case 22:
15             printf("nd"); break;
16         case 3: case 23:
17             printf("rd"); break;
18         default: printf("th"); break;
19     }
20
21     printf(" day of ");
22
23     switch (month) {
24         case 1: printf("January"); break;
25         case 2: printf("February"); break;
26         case 3: printf("March"); break;
27         case 4: printf("April"); break;
28         case 5: printf("May"); break;
29         case 6: printf("June"); break;
30         case 7: printf("July"); break;
31         case 8: printf("August"); break;
32         case 9: printf("September"); break;
33         case 10: printf("October"); break;
34         case 11: printf("November"); break;
35         case 12: printf("December"); break;
36     }
37
38     printf(", 20%.2d.\n", year);
39
40     return 0;
41 }

```

Program 34: Write a program check if the number is automorphic or not.

automorphic.c

```

1  #include<stdio.h>
2  #include<stdlib.h>
3  #include<math.h>
4
5  int main() {
6      int num, sqr, temp, last, n = 0;
7
8      printf("Enter a number: ");
9      scanf("%d", &num);
10     sqr = num * num; //calculating square of num
11     temp = num;
12
13     //Counting number of digits
14     while (temp > 0) {
15         n++;
16         temp = temp / 10;
17     }
18
19     //Extracting last n digits
20     int den = floor(pow(10, n));
21     last = sqr % den;
22
23     if (last == num)
24         printf("Automorphic number \n");
25     else
26         printf("Not Automorphic \n");
27
28     return 0;
29 }
```


Program 35: Write a program show linear search.

linearSearch.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, x, c = 0;
5
6      printf("ENTER SIZE OF ARRAY: ");
7      scanf("%d", &n);
8
9      int a[n];
10
11     for(i=0; i<n; i++) {
12         printf("ENTER ELEMENT %d: ", i + 1);
13         scanf("%d", &a[i]);
14     }
15
16     printf("ENTER ELEMENT TO SEARCH: ");
17     scanf("%d", &x);
18
19     for(i=0; i<n; i++) {
20         if(a[i] == x) {
21             printf("\nFOUND AT INDEX: %d i.e. ELEMENT NUMBER: %d\n", i, i+1);
22             c++;
23             break;
24         }
25     }
26
27     if(c == 0) {
28         printf("\nELEMENT NOT FOUND\n");
29     }
30
31     return 1;
32 }

```

Program 36: Write a program to check if the given number is perfect number.

perfectNumber.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, sum = 0;
5
6      printf("Enter the value of n: ");
7      scanf("%d", &n);
8
9      for(i = 1; i < n; i++) {
10         if (n % i == 0)
11             sum = sum + i;
12     }
13
14     if (n == sum)
15         printf("Entered no. is a perfect no.");
16     else
17         printf("Entered no. is not a perfect no.");
18
19     return 0;
20 }

```


Program 37: Write a program to check if the number is prime number.

primeNumber.c

```

1  #include <stdio.h>
2
3  void main() {
4      int i, n, c = 0;
5
6      printf("Enter a Number : ");
7      scanf("%d", &n);
8
9      for(i = 2; i < n; i++){
10         if(n % i == 0) {
11             printf("NOT a prime number");
12             c++;
13             break;
14         }
15     }
16
17     if(c == 0)
18         printf("%d is a prime number", n);
19 }
```

Program 38: Write a program to print simple multiplication table of a given number.

simpleMultiplicationTable.c

```

1  #include <stdio.h>
2
3  int main() {
4      int num, i = 1;
5
6      printf("Enter a number to calculate table of: \n");
7      scanf("%d", &num);
8
9      for(i = 1; i <= 10; i++) {
10         printf("%d x %d = %d \n", i, num, num * i);
11     }
12
13     return 0;
14 }
```

Program 39: Write a program to reverse a number.

reverseNumber.c

```

1  #include <stdio.h>
2
3  int main() {
4      int num, reverse = 0, remainder;
5
6      printf("Enter a number: ");
7      scanf("%d", &num);
8
9      while (num != 0) {
10         remainder = num % 10;
11         reverse = reverse * 10 + remainder;
12         num = num / 10;
13     }
14
15     printf("Reverse of the number is : %d\n", reverse);
16
17     return 0;
18 }
```

Program 40: Write a program to to reverse a given string.

stringReverse.c

```

1  #include <stdio.h>
2  #include <conio.h>
3
4  void main() {
5      int i, j, k;
6      char str[100];
7      char rev[100];
8
9      printf("Enter a string:\t");
10     scanf("%s", str);
11     printf("The original string is %s\n", str);
12
13     for(i = 0; str[i] != '\0'; i++);
14
15     k = i-1;
16
17     for(j = 0; j <= i-1; j++){
18         rev[j] = str[k];
19         k--;
20     }
21
22     printf("The reverse string is %s\n", rev);
23     getch();
24 }
```

Program 41: Write a program to output * pattern.

starPattern.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, j;
5
6      printf("Enter number of rows : ");
7      scanf("%d", &n);
8
9      for(i = 1; i <= n; i++) {
10         for(j = 1; j <= i; j++) {
11             printf("* ");
12         }
13         printf("\n");
14     }
15
16     return 0;
17 }
```

Program 42: Write a program to output * pattern.

starPattern2.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, j, space;
5
6      printf("Enter number of rows: ");
7      scanf("%d", &n);
8
9      for(i = 1; i <= n; i++) {
10         for(space = 1; space <= n - i; space++) {
11             printf(" ");
12         }
13         for(j = 1; j <= (2 * i - 1); j++) {
14             printf("*");
15         }
16         printf("\n");
17     }
18
19     return 0;
20 }
```

Program 43: Write a program to output number pattern.

numberPattern.c

```

1  #include <stdio.h>
2
3  int main() {
4      int i, j, rows , num = 1;
5
6      printf("Enter number of rows: ");
7      scanf("%d", &rows);
8
9      for(i = 1; i <= rows; i++) {
10         for(j = 1; j <= i; j++) {
11             printf("%d ", num);
12             num++;
13         }
14         printf("\n");
15     }
16
17     return 0;
18 }
```

Program 44: Write a program to output palindromic number pattern.

numberPattern2.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, j, k;
5
6      printf("Enter number of rows: ");
7      scanf("%d", &n);
8
9      for(i = 1; i <= n; i++) {
10         for(j = 1; j <= n-i; j++) {
11             printf(" ");
12         }
13         for(k = i; k >= 1; k--) {
14             printf("%d ", k);
15         }
16         for(k = 2; k <= i; k++) {
17             printf("%d ", k);
18         }
19         printf("\n");
20     }
21
22     return 0;
23 }
```

Program 45: Write a program to output Alphabet Diamond Pattern.

characterPattern.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, j, space;
5
6      printf("Enter number of rows (e.g., 5): ");
7      scanf("%d", &n);
8
9      // Upper half
10     for(i = 1; i <= n; i++) {
11         for(space = 1; space <= n - i; space++) {
12             printf(" ");
13         }
14         char ch = 'A';
15         for(j = 1; j <= i; j++) {
16             printf("%c", ch);
17             ch++;
18         }
19         ch -= 2;
20         for(j = 1; j < i; j++) {
21             printf("%c", ch);
22             ch--;
23         }
24         printf("\n");
25     }
26
27     // Lower half
28     for(i = n - 1; i >= 1; i--) {
29         for(space = 1; space <= n - i; space++) {
30             printf(" ");
31         }
32         char ch = 'A';
33         for(j = 1; j <= i; j++) {
34             printf("%c", ch);
35             ch++;
36         }
37         ch -= 2;
38         for(j = 1; j < i; j++) {
39             printf("%c", ch);
40             ch--;
41         }
42         printf("\n");
43     }
44
45     return 0;
46 }
```

Program 46: Write a program to output Alphabet Pyramid.

characterPattern2.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, j, space;
5
6      printf("Enter number of rows: ");
7      scanf("%d", &n);
8
9      for(i = 1; i <= n; i++) {
10         for(space = 1; space <= n - i; space++) {
11             printf(" ");
12         }
13         char ch = 'A';
14         for(j = 1; j <= i; j++) {
15             printf("%c ", ch);
16             ch++;
17         }
18         printf("\n");
19     }
20
21     return 0;
22 }
```

Program 47: Write a program to output Alphabet–Number Center Pyramid.

characterNumberPattern.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, j, space;
5
6      printf("Enter number of rows: ");
7      scanf("%d", &n);
8
9      for(i = 1; i <= n; i++) {
10         for(space = 1; space <= n - i; space++) {
11             printf(" ");
12         }
13         char ch = 'A';
14         for(j = 1; j <= i; j++) {
15             printf("%c ", ch);
16             ch++;
17         }
18         int num = 1;
19         for(j = 1; j <= i; j++) {
20             printf("%d ", num);
21             num++;
22         }
23         printf("\n");
24     }
25     return 0;
26 }
```


Program 48: Write a program to calculate the sum of given integers.

sum.c

```

1  #include <stdio.h>
2
3  int main() {
4      int n, sum = 0;
5
6      printf("This program sums a series of integers.\n");
7      printf("Enter integers (0 to terminate): ");
8
9      scanf("%d", &n);
10
11     while (n != 0) {
12         sum += n;
13         scanf("%d", &n);
14     }
15
16     printf("The sum is: %d\n", sum);
17
18     return 0;
19 }
```

Program 49: Write a program to calculate the length of the message.

lengthOfMessage.c

```

1  #include <stdio.h>
2
3  int main() {
4      char ch;
5      int len = 0;
6
7      printf("Enter a message: ");
8      ch = getchar();
9
10     while (ch != '\n') {
11         len++;
12         ch = getchar();
13     }
14
15     printf("Your message was %d character(s) long.\n", len);
16
17     return 0;
18 }
```

Program 50: Write a program demonstrate a real-life check book using switch statement.

checkbook.c

```

1  #include <stdio.h>
2
3  int main() {
4      int cmd;
5      float balance = 0.0f, credit, debit;
6
7      printf("*** BANK CHECKBOOK ***\n");
8      printf("Commands: \n1.Clear \n1.Credit \n2.Debit \n");
9      printf("3.Balance \n4.Exit\n\n");
10
11     for (;;) {
12         printf("Enter command: ");
13         scanf("%d", &cmd);
14
15         switch (cmd) {
16             case 1:
17                 balance = 0.0f;
18                 break;
19             case 2:
20                 printf("Enter amount of credit: ");
21                 scanf("%f", &credit);
22                 balance += credit;
23                 break;
24             case 3:
25                 printf("Enter amount of debit: ");
26                 scanf("%f", &debit);
27                 balance -= debit;
28                 break;
29             case 4:
30                 printf("Current balance: $%.2f\n", balance);
31                 break;
32             case 5:
33                 return 0;
34             default:
35                 printf("Commands: \n0=clear \n1=credit \n2=debit \n");
36                 printf("3=balance \n4=exit\n\n");
37                 break;
38         }
39     }
40 }
```

Program 51: Write a program to reverse a series a number.

reverseSeriesOfNumber.c

```

1  #include <stdio.h>
2  #define N 10
3
4  int main() {
5      int a[N], i;
6
7      printf("Enter %d numbers: ", N);
8      for (i = 0; i < N; i++) {
9          scanf("%d", &a[i]);
10     }
11
12     printf("In reverse order:");
13     for (i = N - 1; i >= 0; i--) {
14         printf(" %d", a[i]);
15     }
16     printf("\n");
17
18     return 0;
19 }
```

Program 52: Write a program to check if the number consists of a repeated digit.

repeatedDigits.c

```

1  #include <stdbool.h>
2  #include <stdio.h>
3
4  int main() {
5      bool digit_seen[10] = {false};
6      int digit;
7      long n;
8
9      printf("Enter a number: ");
10     scanf("%ld", &n);
11
12     while (n > 0) {
13         digit = n % 10;
14         if (digit_seen[digit])
15             break;
16         digit_seen[digit] = true;
17         n /= 10;
18     }
19
20     if (n > 0)
21         printf("Repeated digit\n");
22     else
23         printf("No repeated digit\n");
24
25     return 0;
26 }
```

Program 53: Write a program to output a number a random card.

cardsRandom.c

```

1  #include <stdbool.h>
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <time.h>
5
6  #define NUM_SUITS 4
7  #define NUM_RANKS 13
8
9  int main() {
10     bool in_hand[NUM_SUITS][NUM_RANKS] = {false};
11     int num_cards, rank, suit;
12
13     const char rank_code[] = {'2','3','4','5','6','7','8',
14                               '9','T','J','Q','K','A'};
15     const char suit_code[] = {'C','D','H','S'};
16
17     srand((unsigned)time(NULL));
18
19     printf("Enter number of cards in hand: ");
20     scanf("%d", &num_cards);
21
22     printf("Your hand:");
23     while (num_cards > 0) {
24         suit = rand() % NUM_SUITS; /* picks a random suit */
25         rank = rand() % NUM_RANKS; /* picks a random rank */
26         if (!in_hand[suit][rank]) {
27             in_hand[suit][rank] = true;
28             num_cards--;
29             printf("'%c%c' ", rank_code[rank], suit_code[suit]);
30         }
31     }
32     printf("\n");
33
34     return 0;
35 }

```

Program 54: Write a program to print the table of interest according to the years.

interestTable.c

```

1  #include <stdio.h>
2
3  #define NUM_RATES ((int) (sizeof(value) / sizeof(value[0])))
4  #define INITIAL_BALANCE 100.00
5
6  int main(void) {
7      int i, low_rate, num_years, year;
8      double value[5];
9
10     printf("Enter interest rate: ");
11     scanf("%d", &low_rate);
12     printf("Enter number of years: ");
13     scanf("%d", &num_years);
14
15     printf("\nYears");
16     for (i = 0; i < NUM_RATES; i++) {
17         printf("%6d%%", low_rate + i);
18         value[i] = INITIAL_BALANCE;
19     }
20     printf("\n");
21
22     for (year = 1; year <= num_years; year++) {
23         printf("%3d", year);
24         for (i = 0; i < NUM_RATES; i++) {
25             value[i] += (low_rate + i) / 100.0 * value[i];
26             printf("%7.2f", value[i]);
27         }
28         printf("\n");
29     }
30
31     return 0;
32 }
```

Program 55: Write a program to print pun using function.

punFunction.c

```

1  #include <stdio.h>
2
3  void print_pun() {
4      printf("To C, or not to C: that is the question.\n");
5  }
6
7  int main(void) {
8      print_pun();
9      return 0;
10 }
```

Program 56: Write a program to give average of the given number.

average.c

```

1  #include <stdio.h>
2
3  double average(double a, double b) {
4      return (a + b) / 2;
5  }
6
7  int main() {
8      double x, y, z;
9
10     printf("Enter three numbers: ");
11     scanf("%lf%lf%lf", &x, &y, &z);
12
13     printf("Average of %g and %g: %g\n", x, y, average(x, y));
14     printf("Average of %g and %g: %g\n", y, z, average(y, z));
15     printf("Average of %g and %g: %g\n", x, z, average(x, z));
16
17     return 0;
18 }
```

Program 57: Write a program to give countdown of the user given number.

countDown.c

```

1  #include <stdio.h>
2
3  void printCount(int n) {
4      printf("T minus %d and counting\n", n);
5  }
6
7  int main() {
8      int i, n;
9
10     printf("Enter the number to start countdown from: ");
11     scanf("%d", &n);
12
13     for (i = n; i > 0; --i) {
14         printCount(i);
15     }
16
17     return 0;
18 }
```


Program 58: Write a program check if the number is prime using a function.

primeFunction.c

```

1  #include <stdbool.h>
2  #include <stdio.h>
3
4  bool isPrime(int n) {
5      int divisor;
6
7      if (n <= 1)
8          return false;
9
10     for (divisor = 2; divisor * divisor <= n; divisor++) {
11         if (n % divisor == 0)
12             return false;
13     }
14
15     return true;
16 }
17
18 int main() {
19     int n;
20
21     printf("Enter a number: ");
22     scanf("%d", &n);
23
24     if (isPrime(n))
25         printf("Prime\n");
26     else
27         printf("Not prime\n");
28
29     return 0;
30 }

```

Program 59: Write a program to give factorial of a given number.

factorial.c

```

1  #include <stdio.h>
2
3  // Function to calculate factorial
4  int factorial(int n) {
5      int fact = 1;
6
7      for(int i = 1; i <= n; i++) {
8          fact *= i; // multiply fact by each number up to n
9      }
10
11     return fact;
12 }
13
14 int main() {
15     int num;
16
17     printf("Enter a number: ");
18     scanf("%d", &num);
19
20     printf("Factorial of %d = %d", num, factorial(num));
21
22     return 0;
23 }

```

Program 60: Write a program to check if the number is Armstrong number.

armstrongNumber.c

```

1  #include <stdio.h>
2  #include <math.h>
3
4  int isArmstrong(int n) {
5      int temp = n, sum = 0, digits = 0;
6
7      // Count digits
8      while(temp != 0) {
9          digits++;
10         temp /= 10;
11     }
12
13     temp = n;
14     while(temp != 0) {
15         int r = temp % 10;
16         sum += pow(r, digits); // r^digits
17         temp /= 10;
18     }
19
20     if(sum == n)
21         return 1; // Armstrong
22     else
23         return 0; // Not Armstrong
24 }
25
26 int main() {
27     int num;
28
29     printf("Enter a number: ");
30     scanf("%d", &num);
31
32     if (isArmstrong(num))
33         printf("%d is an Armstrong number", num);
34     else
35         printf("%d is NOT an Armstrong number", num);
36
37     return 0;
38 }

```

Program 61: Write a program to give Fibonacci series.

fibonnaciSeries.c

```
1  #include <stdio.h>
2
3  // Function to print Fibonacci series
4  void fibonacci(int n) {
5      int a = 0, b = 1, c;
6
7      printf("Fibonacci Series: %d %d ", a, b);
8
9      for(int i = 3; i <= n; i++) {
10         c = a + b;
11         printf("%d ", c);
12         a = b;
13         b = c;
14     }
15 }
16
17 int main() {
18     int n;
19
20     printf("Enter number of terms: ");
21     scanf("%d", &n);
22
23     fibonacci(n);
24     return 0;
25 }
```

Program 62: Write a program to demonstrate quick sort algorithm.

quickSort.c

```

1  /* Sorts an array of integers using Quicksort algorithm */
2  #include <stdio.h>
3
4  #define N 10
5
6  void quicksort(int a[], int low, int high);
7  int split(int a[], int low, int high);
8
9  int main() {
10     int a[N], i;
11
12     printf("Enter %d numbers to be sorted: ", N);
13     for (i = 0; i < N; i++) {
14         scanf("%d", &a[i]);
15     }
16
17     quicksort(a, 0, N - 1);
18
19     printf("In sorted order: ");
20     for (i = 0; i < N; i++){
21         printf("%d ", a[i]);
22     }
23     printf("\n");
24
25     return 0;
26 }
27
28 void quicksort(int a[], int low, int high) {
29     int middle;
30
31     if (low >= high) return;
32
33     middle = split(a, low, high);
34     quicksort(a, low, middle - 1);
35     quicksort(a, middle + 1, high);
36 }
37
38 int split(int a[], int low, int high) {
39     int part_element = a[low];
40
41     for (;;) {
42         while (low < high && part_element <= a[high]) {
43             high--;
44             int split::part_element
45         }
46         if (low >= high) break;
47
48         a[low++] = a[high];
49
50         while (low < high && a[low] <= part_element){
51             low++;
52         }
53         if (low >= high) break;
54
55         a[high--] = a[low];
56     }
57
58     a[high] = part_element;
59     return high;
60 }

```

Programs 63: Write a program to implement simple banking system**bankingSystem.c**

```

1  #include <stdio.h>
2
3  // Structure to store account information
4  struct BankAccount {
5      char name[50];
6      float balance;
7  };
8
9  // Function to deposit money
10 void deposit(struct BankAccount *acc) {
11     float amount;
12
13     printf("Enter amount to deposit: ");
14     scanf("%f", &amount);
15
16     acc->balance += amount;
17
18     printf("%.2f deposited successfully.\n", amount);
19 }
20
21 // Function to withdraw money
22 void withdraw(struct BankAccount *acc) {
23     float amount;
24
25     printf("Enter amount to withdraw: ");
26     scanf("%f", &amount);
27
28     if(amount <= acc->balance) {
29         acc->balance -= amount;
30         printf("%.2f withdrawn successfully.\n", amount);
31     } else {
32         printf("Insufficient Balance!\n");
33     }
34 }
35
36 // Function to display account details
37 void showBalance(struct BankAccount acc) {
38     printf("\nAccount Holder: %s\n", acc.name);
39     printf("Current Balance: %.2f\n", acc.balance);
40 }
41
42 int main() {
43     struct BankAccount customer;
44     int choice;
45
46     printf("Enter Account Holder Name: ");
47     scanf("%s", customer.name);
48
49     customer.balance = 1000;
50
51     do {
52         printf("\n===== BANK MENU =====\n");
53         printf("1. Deposit\n");
54         printf("2. Withdraw\n");
55         printf("3. Check Balance\n");
56         printf("4. Exit\n");
57
58         printf("Enter Choice: ");
59         scanf("%d", &choice);
60
61

```

```
62     switch(choice) {
63         case 1:
64             deposit(&customer);
65             break;
66         case 2:
67             withdraw(&customer);
68             break;
69         case 3:
70             showBalance(customer);
71             break;
72         case 4:
73             printf("Thank You For Banking With Us.\n");
74             break;
75         default:
76             printf("Invalid Choice!\n");
77     }
78     while(choice != 4);
79
80     return 0;
81 }
```


Programs 64: Write a program to implement TicTacToe Game**ticTacToe.c**

```

10 #include <stdio.h>
11 #include <conio.h> // Optional: for getch() to pause screen
12 #include <stdlib.h> // For system("cls") to clear screen
13
14 // Global 3x3 array to store the game board
15 char board[3][3] = {
16     {'1', '2', '3'},
17     {'4', '5', '6'},
18     {'7', '8', '9'}
19 };
20
21 // Function Prototypes
22 void displayBoard();
23 int checkWinner();
24 int isDraw();
25 void markBoard(int choice, char mark);
26
27 // Main Function
28 int main() {
29     int player = 1; // Player 1 starts first
30     int choice; // Position selected by player
31     char mark; // X or O
32     int status = -1; // -1 = game continues, 1 = win, 0 = draw
33
34     do {
35         system("cls"); // Clear screen (for better visibility)
36
37         printf("\n\tTIC TAC TOE GAME\n");
38         printf("\t-----\n");
39         printf("\nPlayer 1 (X) - Player 2 (O)\n\n");
40
41         // Display current game board
42         displayBoard();
43
44         // Determine which player's turn it is
45         player = (player % 2) ? 1 : 2;
46         mark = (player == 1) ? 'X' : 'O';
47
48         // Ask the player for a move
49         printf("Player %d, enter the position (1-9): ", player);
50         scanf("%d", &choice);
51
52         // Update board with the player's mark
53         markBoard(choice, mark);
54
55         // Check if someone has won or if it's a draw
56         if (checkWinner()) {
57             system("cls");
58             printf("\n\tTIC TAC TOE GAME\n");
59             printf("\t-----\n");
60             printf("\nPlayer 1 (X) - Player 2 (O)\n\n");
61             displayBoard();
62             printf("\n==> Player %d wins! \n", player);
63             status = 1; // Game ends with a win
64             break;
65         } else if (isDraw()) {
66             system("cls");
67             printf("\n\tTIC TAC TOE GAME\n");
68             printf("\t-----\n");
69             printf("\nPlayer 1 (X) - Player 2 (O)\n\n");
70             displayBoard();
71             printf("\n==> Game Draw! \n");
72             status = 0; // Game ends with a draw
73             break;
74         }
75
76         player++; // Move to next player
77
78     } while (status == -1);
79

```

```

80     printf("\nThank you for playing Tic Tac Toe!\n");
81
82     getch(); // Wait for key press before closing
83     return 0;
84 }
85
86 /*-----
87     Function : displayBoard()
88     Purpose  : Displays the current Tic Tac Toe board
89     -----*/
90 void displayBoard() {
91     printf("\n\t %c | %c | %c ", board[0][0], board[0][1], board[0][2]);
92     printf("\n\t---|---|---");
93     printf("\n\t %c | %c | %c ", board[1][0], board[1][1], board[1][2]);
94     printf("\n\t---|---|---");
95     printf("\n\t %c | %c | %c \n\n", board[2][0], board[2][1], board[2][2]);
96 }
97
98 /*-----
99     Function : markBoard()
100    Purpose  : Places player mark on selected position
101    -----*/
102 void markBoard(int choice, char mark) {
103     int row = (choice - 1) / 3; // Convert 1-9 choice into 2D indices
104     int col = (choice - 1) % 3;
105
106     // Check if position is valid and not already taken
107     if (choice < 1 || choice > 9 || board[row][col] == 'X' || board[row][col] == 'O') {
108         printf("Invalid move! Try again.\n");
109         getch();
110     } else {
111         board[row][col] = mark;
112     }
113 }
114
115 /*-----
116     Function : checkWinner()
117     Purpose  : Checks all possible winning combinations
118     -----*/
119 int checkWinner() {
120     // Check rows and columns
121     for (int i = 0; i < 3; i++) {
122         if ((board[i][0] == board[i][1] && board[i][1] == board[i][2]) ||
123             (board[0][i] == board[1][i] && board[1][i] == board[2][i])) {
124             return 1;
125         }
126     }
127
128     // Check diagonals
129     if ((board[0][0] == board[1][1] && board[1][1] == board[2][2]) ||
130         (board[0][2] == board[1][1] && board[1][1] == board[2][0])) {
131         return 1;
132     }
133
134     return 0; // No winner yet
135 }
136
137 /*-----
138     Function : isDraw()
139     Purpose  : Checks if all cells are filled (draw)
140     -----*/
141 int isDraw() {
142     for (int i = 0; i < 3; i++) {
143         for (int j = 0; j < 3; j++) {
144             if (board[i][j] != 'X' && board[i][j] != 'O')
145                 return 0; // Still empty cell, game continues
146         }
147     }
148     return 1; // All cells filled, game is a draw
149 }

```

Programs 65: Write a program to implement simple library system**librarySystem.c**

```

1  #include <stdio.h>
2  #include <string.h>
3
4  #define MAX_BOOKS 100
5
6  // Structure to store book information
7  struct Book{
8      int id;
9      char title[50];
10     char author[50];
11     int available;
12 };
13
14 struct Book library[MAX_BOOKS];
15 int count = 0;
16
17 // Function: Add Book
18 void addBook() {
19     printf("\nEnter Book ID: ");
20     scanf("%d", &library[count].id);
21
22     printf("Enter Book Title: ");
23     scanf(" %[^\\n]", library[count].title);
24
25     printf("Enter Author Name: ");
26     scanf(" %[^\\n]", library[count].author);
27
28     library[count].available = 1;
29     count++;
30
31     printf("\nBook Added Successfully!\\n");
32 }
33
34 // Function: Display Books
35 void displayBooks() {
36     int i;
37
38     if(count == 0) {
39         printf("\\nNo Books Available.\\n");
40         return;
41     }
42
43     printf("\\n-----\\n");
44
45     for(i=0;i<count;i++) {
46         printf("Book ID   : %d\\n", library[i].id);
47         printf("Title      : %s\\n", library[i].title);
48         printf("Author    : %s\\n", library[i].author);
49
50         if(library[i].available)
51             printf("Status    : Available\\n");
52         else
53             printf("Status    : Issued\\n");
54
55         printf("\\n-----\\n");
56     }
57 }
58
59 // Function: Search Book
60 void searchBook(){
61     int id, i;
62
63     printf("Enter Book ID to Search: ");
64     scanf("%d",&id);
65
66     for(i=0;i<count;i++) {
67         if(library[i].id == id) {
68             printf("\\nBook Found!\\n");
69
70             printf("Book ID : %d\\n", library[i].id);
71             printf("Title   : %s\\n", library[i].title);
72             printf("Author  : %s\\n", library[i].author);
73
74             return;
75         }

```

```

76     }
77
78     printf("Book Not Found!\n");
79 }
80
81 // Function: Issue Book
82 void issueBook() {
83     int id, i;
84
85     printf("Enter Book ID to Issue: ");
86     scanf("%d", &id);
87
88     for(i=0; i<count; i++) {
89         if(library[i].id == id) {
90             if(library[i].available) {
91                 library[i].available = 0;
92                 printf("Book Issued Successfully!\n");
93             } else {
94                 printf("Book Already Issued!\n");
95             }
96         }
97     }
98     return;
99 }
100
101 printf("Book Not Found!\n");
102 }
103
104 // Function: Return Book
105 void returnBook() {
106     int id, i;
107
108     printf("Enter Book ID to Return: ");
109     scanf("%d", &id);
110
111     for(i=0; i<count; i++) {
112         if(library[i].id == id) {
113             library[i].available = 1;
114             printf("Book Returned Successfully!\n");
115         }
116     }
117     return;
118 }
119
120 printf("Book Not Found!\n");
121 }
122
123 // Main Function
124 int main() {
125     int choice;
126
127     do {
128         printf("\n===== LIBRARY MANAGEMENT =====\n");
129         printf("1. Add Book\n");
130         printf("2. Display Books\n");
131         printf("3. Search Book\n");
132         printf("4. Issue Book\n");
133         printf("5. Return Book\n");
134         printf("6. Exit\n");
135
136         printf("Enter Choice: ");
137         scanf("%d", &choice);
138

```

```
139
140     switch(choice) {
141         case 1:
142             addBook();
143             break;
144         case 2:
145             displayBooks();
146             break;
147         case 3:
148             searchBook();
149             break;
150         case 4:
151             issueBook();
152             break;
153         case 5:
154             returnBook();
155             break;
156         case 6:
157             printf("Thank You!\n");
158             break;
159         default:
160             printf("Invalid Choice!\n");
161     }
162
163     } while(choice != 6);
164
165     return 0;
166 }
```